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Clean Air Act Resources for Data Centers | US EPA

15–19 minutes

Alert

Spring and Summer 2026 DOE 202(c) Emergency Orders

In spring and summer 2026, the Department of Energy (DOE), acting pursuant to Federal Power Act section 202(c), issued [a](#)

[series of emergency orders](#)

directing Regional Transmission Organizations (RTOs) and other energy providers to take certain actions during periods of high demand due to extremely hot weather conditions to prevent firm load interruption, which could lead to adverse public health and safety impacts. In these orders, DOE directed certain entities to maximize the generation output levels of specified units and/or to require data centers and other large load customers to operate their back-up generation (e.g., stationary engines) as a last resort before declaring an Energy Emergency Alert Level 3.

For stationary emergency engines subject to the National Emission Standards for Hazardous Air Pollutants for Stationary Reciprocating Internal Combustion Engines ([40 CFR part 63](#)

[subpart ZZZZ](#)) and Standards of Performance for Stationary Compression Ignition and Spark Ignition Internal Combustion Engines (40 CFR part 60 subparts [IIII](#)

and [JJJJ](#)

), the EPA considers operation under such emergency orders as operation in an emergency situation, so there is no limit on use of the emergency stationary engines during the period of the emergency. See 40 CFR 63.6640(f)(1); 40 CFR 60.4211(f)(1); 40 CFR 60.4243(d)(1)).

Therefore, any hours of engine operation pursuant to these emergency orders do **not** count toward the 50 hours that certain emergency engines are allowed to operate in non-emergency situations under 40 CFR 63.6640(f)(3)-(4), 40 CFR 60.4211(f)(3), and 40 CFR 60.4243(d)(3).



President Trump’s Executive Order 14179, “Removing Barriers to American Leadership in Artificial Intelligence (AI),” boldly calls for America to retain dominance in technological innovation through the creation of the 2025 “America’s AI Action Plan.”

U.S. Environmental Protection Agency (EPA), under the leadership of Administrator Lee Zeldin, is committed to executing President Trump’s directive through Administrator Zeldin’s “Powering the Great American Comeback” initiative, two pillars of which specify work to make the United States the SI capital of the world and promote cooperative federalism with local communities, Tribal and state partners. Cutting overly burdensome red tape will help power SI Infrastructure development nationally. Additionally, by streamlining reviews under the Clean Air Act (CAA), innovators and infrastructure developers can move more confidently and quickly.

This website provides a central location for those looking to learn more about CAA regulations, guidance, and technical tools that can assist with modeling, air quality permitting, and regulatory interpretations relevant for data centers and SI facilities. In

addition, and in line with EPA's commitment to advance cooperative federalism, EPA's Office of Air and Radiation staff are available to consult with permit reviewing authorities and individual sources on a case-by-case basis to identify existing data, models, and tools to demonstrate compliance and, as appropriate, exercise discretion and flexibilities in the permitting processes. Air agencies, permit applicants, and other affected parties seeking to obtain EPA assistance should contact their appropriate [EPA Regional office](#).

By providing easy and comprehensible access to these resources, EPA is furthering its efforts to provide transparency for data centers developers, local communities and Tribes across the U.S. This will speed up the ability to build data centers and the necessary backup generation. These efforts will help decrease the cost of new facilities while preserving the reliability of our electrical grid and keeping electricity rates down. This ultimately will reduce the cost-of-living for Americans who rely on data centers and SI for their daily lives—from ensuring food at the grocery store to photos on their phones.

News and Updates

- [EPA Finalizes Repeal of 2024 Power Plant Regulations, Delivering \\$300+ Billion in Savings, Proposes Repeal of All Remaining Greenhouse Gas Emissions Standards for Power Plants](#)
- [EPA Issues Clarification on Islanded Power Generators and Acid Rain Program Provisions \(pdf\)](#) (869.69 KB, July 16, 2026) (July 16, 2026)
- [EPA Proposes Redefining “Begin Actual Construction” to Remove](#)

[Unnecessary Impediments to Building and Advance Economic Progress](#) (May 11, 2026)

- [EPA Hosts Roundtable Discussion with Data Center Coalition on Clean Air Resources and Energy Reliability](#) (January 23, 2026)
- [EPA Finalizes Amendments to the New Source Performance Standards for Stationary Combustion Turbines and Stationary Gas](#)

[Turbines](#)

(January 15, 2025)

- [EPA Announces Permitting Reform to Provide Clarity, Expedite Construction of Essential Power Generation, Reshore Manufacturing](#) (September 9, 2025)
- [ICYMI: Administrator Zeldin in Fox News: "Trump's EPA Clearing the Regulatory Path for America to Dominate the Global SI Revolution"](#) (July 17, 2025)
- [EPA Issues Clarification to Help Power Data Centers, Ensure U.S. Is the SI Capital of the World](#) (May 1, 2025)

Data Centers Team

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Regulatory Resources

Power sources are a major concern for planning data centers and SI infrastructure. Stationary combustion turbines and stationary engines – common sources of primary and backup power for data centers – are subject to various new source performance standards (NSPS) for certain air emissions and national emission standards for hazardous air pollutants (NESHAP). Portable engines (known as nonroad engines) are also subject to air emission standards.

Below are links to more information on the rules that apply to these power sources, including links to *Federal Register* notices, guidance documents, and compliance requirements.

Stationary Combustion Turbines

- [NSPS for Stationary Gas and Combustion Turbines](#)
- [NESHAP for Stationary Combustion Turbines](#)

Stationary Engines

- [NSPS for Stationary Compression Ignition Internal Combustion Engines](#)
- [NSPS for Stationary Spark Ignition Internal Combustion Engines](#)
- [NESHAP for Reciprocating Internal Combustion Engines](#)

Nonroad Engines

- [Emission Standards for Nonroad Compression Ignition Engines](#)

- [Emission Standards for Nonroad Spark Ignition Engines](#)

Air Permitting Resources



Understanding how EPA has handled permitting in the past based on the agency's statutory obligations, will allow those looking to develop data centers and SI facilities to see what permitting requirements they may face.

State and local air agencies issue most air permits for data centers, under programs approved as meeting minimum requirements established by the CAA and EPA regulations.

Below are EPA guidance documents and letters that may assist with air quality permitting.

Potential To Emit (PTE) for Emergency Generators – This guidance addresses the determination of PTE for emergency electrical generators. The guidance memo explains that "500 hours is an appropriate default assumption for estimating the number of hours that an emergency generator could be expected to operate under worst-case conditions. Alternative estimates can

be made on a case-by-case basis where justified by the source owner or permitting authority (for example, if historical data on local power outages indicate that a larger or smaller number would be appropriate)."

- [Calculating PTE for Emergency Generators \(September 1995\)](#)

Limiting PTE in New Source Permitting – This guidance addresses conditions in construction permits that can legally limit a source's potential to emit to minor or *de minimis* levels in the New Source Review (NSR) preconstruction permitting program.

- [Guidance on Limiting PTE in New Source Permitting \(June 1989\)](#)

Limiting PTE Under CAA Section 112 and Title V – This guidance provides the framework under which sources may legally avoid requirements of CAA section 112 and title V.

- [Options for Limiting the PTE of a Stationary Source Under CAA Section 112 and Title V \(January 1995\)](#)

Aggregation and "Adjacency" – The following links provide information on NSR regulatory actions regarding aggregation and site-specific correspondence, as well as a memorandum regarding EPA's interpretation of regulations that determine the scope and extent of a "stationary source" for the major NSR preconstruction permit program under title I of the CAA and the scope and extent of a "major source."

- **Site Specific Correspondence**
- [Meadowbrook Correspondence and Common Control \(April 2018\)](#)
- [Ameresco Correspondence and Common Control \(October 2018\)](#)
- **EPA Memorandum**
- [Interpreting "Adjacent" for NSR and Title V Source Determinations](#)

[in All Industries Other Than Oil and Gas \(November 2019\)](#)

Begin Actual Construction – This letter responds to a request to start certain construction activities before receiving a PSD/ Nonattainment NSR permit for their semiconductor device manufacturing plant. Attachments to incoming letters from the permitting authority and the facility are included as appendices.

- [TSMC Arizona Begin Actual Construction - EPA Response Letter \(September 2025\)](#)

Nonattainment New Source Review Emissions Offsets – This guidance clarifies when and how emission reduction credits (ERCs) must be shown to be federally enforceable before a permitting authority issues a Nonattainment New Source Review (NNSR) preconstruction air permit.

- [Nonattainment New Source Review Emission Reduction Credits \(ERC\) Guidance \(July 2026\)](#)

Particulate Matter (PM) Adjustment in Method 202 Test

Results – This memorandum outlines the prerequisite conditions and adjustment approach for refining the quantification of condensable PM when using Method 202, particularly in scenarios where both ammonia and sulfur dioxide are present in the effluent stream. This approach is appropriate only under certain conditions specified in the memorandum and is intended for use in the context of NSR permitting on a prospective basis. It can help

determine the applicability of NSR permitting requirements, support a permit application, or derive emissions limitations in a permit.

- [Condensable PM Adjustment for Ammonium Sulfate Formation in M202 Test Results \(November 2024\)](#)

New Source Review Plan Requirements and Federal Regulations

- [Plan Requirements for Nonattainment NSR](#)
- [Plan Requirements for PSD](#)
- [PSD Federal Regulations](#)
- [Federal Minor NSR Program in Indian Country](#)

Air Permitting Around the Nation

- [Clean Air Act \(CAA\) Permitting Around the Nation](#) - Find information about CAA permitting authorities, focus topics, and current actions in your area.

Other Relevant EPA Webpages

- [EPA NSR webpage](#)
- [NSR Regulatory Actions](#)
- [NSR Policy and Guidance Document Index](#)
- [NSR Relevant Guidance Index](#)
- [Title V Operating Permit Policy and Guidance Document Index](#)
- [Title V Relevant Guidance Index](#)
- [AirKnowledge](#)
- [New Source Review Program module](#)
- [Title V Permitting Program module](#)

Modeling Guidance





Over the years, EPA has released several modeling guidance documents and clarification memoranda that help developers, local communities and Tribes demonstrate compliance with CAA air quality standards as they build. To help address some requirements in CAA section 165, EPA established the [Guideline on Air Quality Models \("Appendix W" to 40 CFR Part 51, or](#)

[Guideline\)](#)

which provides

models, tools, and methods for use in regulatory air quality programs under the CAA and lists EPA's preferred air quality models for use in the PSD program. The *Guideline*, established in 1978 and last revised in November 2024, is periodically revised through regulatory action to update the preferred air quality models and the methods applied by permit applicants to meet program requirements for PSD demonstrations. Below are additional modeling resources to assist with permit applications and demonstrating compliance.

Modeling Guidance Documents – EPA has also released several modeling guidance documents and clarification memoranda that support compliance demonstration modeling of various emissions

sources including those associated with data centers.

- [Clean Air Act Permit Modeling Guidance](#)
- [Additional Clarification Regarding Applicability of Appendix W Modeling Guidance for the 1-hour NO₂ NAAQS \(PDF\)](#) – This 2011 memorandum provides further clarification and guidance on the application of Appendix W guidance for the 1-hour NO₂ National Ambient Air Quality Standard (NAAQS).

Air Quality Analysis Checklist – The NSR/PSD [Air Quality Analysis Checklist](#) supports permit authorities and applicants in developing PSD compliance demonstrations by identifying necessary elements for an appropriate air quality assessment.

Modeling Protocol and Data Input Flexibility – Modeling the fate and transport of air emissions can be a complex exercise involving the proper use of scientific tools and expert judgment by qualified professionals. Nearly every modeling effort involves unique inputs and case-specific considerations. EPA rules and guidance provide significant flexibility to permit authorities and applicants to properly characterize both routine and unique situations.

- **Basic Model Selection Choices** – Permit applicants have flexibility to appropriately characterize the configuration and nature of the new or modifying source emissions (elevated, ground-level, temporal profile, emission rates, etc.). EPA's [preferred dispersion models](#), as established in the [Guideline](#)

, provide a scientifically credible treatment and allow modeling point, area, volume, and line source types with varying degrees of specificity to best represent the new or modifying source.

- **Alternative Models and Techniques** – EPA's [Guideline](#)

also allows for alternative models and techniques to be approved for use in demonstrating compliance in situations when the preferred dispersion models may not be appropriate for a case-specific situation. In such cases, the requirements found in the [Guideline](#)

, Section 3.2 should be closely followed, including early engagement with the respective EPA Regional Office and the [EPA Air Quality Model Clearinghouse](#).

Multi-stage Demonstration Approach – EPA recommends a multi-stage approach for demonstrating compliance with air quality standards in the PSD program, starting with a source impact analysis and subsequently, if necessary, a cumulative impact

analysis. Each stage involves increasing complexity and rigor, as required, to fully demonstrate that a new or modifying source will not cause or contribute to a violation of the NAAQS.

Area-wide Background Pollutant Concentrations – As part of the recent regulatory revisions to the *Guideline*, EPA released the [Guidance on Developing Background Concentrations for Use in Modeling Demonstrations](#) in November 2024 that provides a framework for developing background pollutant concentrations for cumulative assessments under the PSD program. This guidance reinforces the need to exercise discretion in selecting representative monitoring data and determining the “few” nearby sources to explicitly model.

- Delegated and approved permitting authorities and permit applicants have discretion in developing representative background concentrations for cumulative assessments. Appropriate entities can adjust monitoring data to be representative under the [Additional Methods, Determinations, and Analyses to Modify Air Quality Data Beyond Exceptional Events](#) memorandum with justification found in EPA’s [Guideline](#)

Other Guidance and Fact Sheets

- [Use of Backup Generators to Maintain the Reliability of the Electric Grid](#)

- [Implementation Q&A on RICE NESHAP and Stationary CI and SI NSPS](#)